



COMPUTER SCIENCE

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Software Developer · Age 22

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Selected work

1. Cosmic Observer
Windows astronomy desktop app — 3D sky, NASA data, AI tutor.

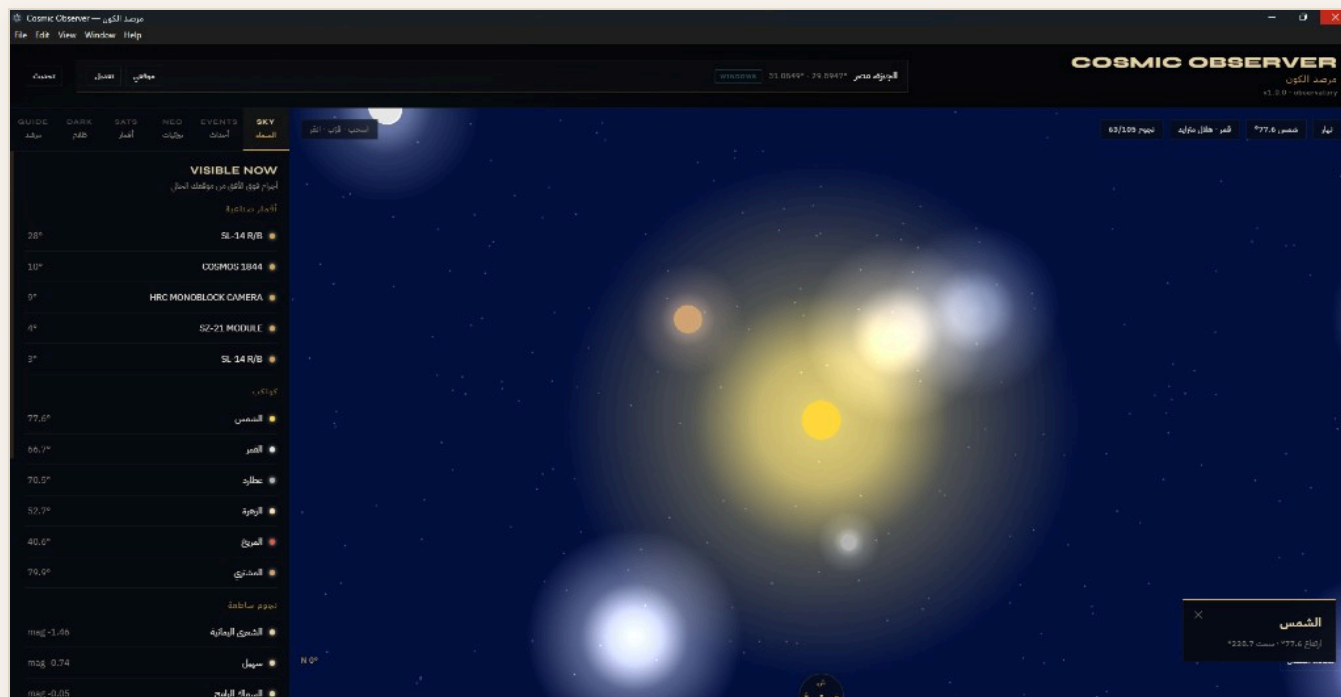
2. Vertical Farm Analysis
Python pipeline — telemetry to charts and PDF report.

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I build desktop applications and data pipelines with real deliverables: demos, screenshots, and client-ready reports.

Cosmic Observer

Windows desktop astronomy app with interactive 3D sky, NEO tracking, orbital debris, dark-matter research views, and an AI tutor — shipped as a native **.exe**.



Capabilities

- Live 3D sky from GPS or manual location
- Satellites, planets, and bright stars
- Sun / Moon status and sky events
- Modules: Sky, Events, NEO, Sats, Dark, Guide
- Arabic + English interface

Links

- github.com/engboda6/cosmic-observer
- License: MIT
- Platform: Windows (.exe)
- Demo video on portfolio site

Stack & Features

JavaScript end to end — React + Three.js for the sky, Node.js/Express for APIs, Electron for the Windows desktop shell.

JavaScript

React 18

Three.js

Vite

Node.js

Express

astronomy-engine

OpenAI

Electron 33

GLSL

Features

- Interactive 3D celestial sphere
- Sunrise / sunset and moon phases
- NASA NeoWs near-Earth objects
- CelesTrak debris / TLE tracking
- Dark-matter lensing research views
- Optional OpenAI astronomy tutor

Data sources

- NASA NeoWs
- JPL Sentry / CAD
- NASA CNEOS Fireball
- CelesTrak (TLE)
- VizieR / CDS

Structure

- electron/ — desktop shell
- server/ — local APIs
- src/ — React + Three.js

Vertical Farm Analysis

Pure Python pipeline for 600 telemetry rows. Report prepared by **Abd Alrahman Ahmed**.

- 1

prepare_dataset.py — clean CSV and parse Stack / Rack / Row IDs
- 2

analysis.py — decay matrix for 60 channels, harvest index, feeding balances
- 3

charts.py — EC distribution, heatmap, harvest charts
- 4

report.py — three-page PDF for delivery
- 5

run_pipeline.py — full run in one command

Vertical Farm Nutrient Solution & Biomass Analysis

Page 1 - Executive Summary & Key Findings

Prepared by Abd Alrahman Ahmed | 15 Jul 2020

This report analyses 600 production rows of real agricultural automation telemetry and water-chemistry logs across the vertical farm (6 stacks x 10 racks x 10 rows = 60 irrigation channels). The objective is to map where nutrient-solution EC decays, where plant growth stalls, and how circulation, light-cycle hours and humidity drive vertical-stack yields and operating cost.

Key metrics

Production rows analysed	600
Irrigation channels (Stack-Rack)	60
Mean final EC	1.69 mS/cm
Rows within optimal EC window	92%
Mean EC decay per cycle	15.3%
Rows with stalled growth	8 (1%)
Low-circulation channels	12 of 60
HIGH-priority decay channels	12
Total harvest yield	7,923 kg

Key findings:

- Water circulation is the dominant lever: channels with low flow show markedly faster EC decay and a higher share of stalled growth.
- Faster nutrient decay drives EC away from the optimal window, which is a leading indicator of stalled growth and lost yield.
- Yield loss is not uniform - it clusters in a small set of channels that can be fixed with targeted maintenance rather than facility-wide change.
- Worst channel is D-R07 (score 0.88, 19% decay, 11.0 L/min).

Page 1 — Summary

Vertical Farm Nutrient Solution & Biomass Analysis

Page 2 - Nutrient Solution Decay Acceleration Matrix

Prepared by Abd Alrahman Ahmed | 15 Jul 2020

The decay acceleration matrix aggregates telemetry per irrigation channel (Stack-Rack) and scores each 0-1 by combining nutrient EC decay (40%), circulation deficit (30%), EC-drop magnitude (15%) and growth-stall rate (10%). The grid below shows all 60 channels; hot (red) cells are critical and should be serviced first.

Top decay-acceleration channels (worst 10)

Rank	Channel	Circ L/min	Decay%	Final EC	Harvest kg	Score	Prio
1	D-R07	11.0	18.9	1.66	12.4	0.88	HIGH
2	E-R07	12.6	19.0	1.70	14.1	0.84	HIGH
3	D-R01	12.5	18.2	1.59	14.5	0.75	HIGH
4	F-R05	12.2	17.8	1.72	12.6	0.75	HIGH
5	A-R04	12.3	17.6	1.62	11.7	0.71	HIGH
6	D-R05	12.1	16.9	1.61	11.5	0.66	HIGH
7	E-R01	12.6	16.7	1.66	11.9	0.62	HIGH
8	C-R02	11.4	15.5	1.62	13.6	0.61	HIGH
9	C-R03	11.7	16.9	1.56	15.1	0.60	HIGH
10	C-R09	11.2	16.1	1.71	11.8	0.59	HIGH

Page 2 — Matrix

Vertical Farm Nutrient Solution & Biomass Analysis

Page 3 - Harvest Index & Optimal Nutrient Feeding B

Prepared by Abd Alrahman Ahmed | 15 Jul 2020

Average Harvest Weight Index by crop variety

Average harvest weight index by crop variety

Crop variety	Rows	Avg kg	Median kg	Final EC	Index
Tuscan Kale	95	18.91	18.87	1.70	143.2
Butterhead Lettuce	110	15.02	14.97	1.66	113.8
Red Oak Leaf Lettuce	95	13.54	13.56	1.70	102.6
Genovese Basil	93	11.14	11.10	1.67	84.3
Arugula	94	10.77	10.74	1.72	81.6
Baby Spinach	113	10.08	10.11	1.68	78.3

Optimal nutrient feeding balances (derived from best-yielding rows)

For each crop, targets are the average conditions of its top-quartile (non-stalled) rows.

Crop variety	Target EC	Init EC	Light h	Humid %	Decays	Exp. kg
Tuscan Kale	1.65	1.99	16.0	66	17.8	20.58
Butterhead Lettuce	1.64	1.93	15.8	65	15.0	16.25
Red Oak Leaf Lettuce	1.71	2.00	15.8	65	14.5	14.78
Genovese Basil	1.73	2.06	16.0	66	14.3	15.16

Page 3 — Harvest

Libraries & Contact

Raw farm CSV becomes a clean dataset, analysis tables, charts, Excel files, and a three-page report.

Libraries

- **pandas** — cleaning & aggregation
- **numpy** — calculations
- **matplotlib** — charts & PDF figures
- **openpyxl** — Excel export

Deliverables

- Clean structured dataset
- Analysis for 60 irrigation channels
- Charts and Excel workbooks
- Three-page nutrient PDF

Contact

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